SI 140A.02 Probability & Statistics for EECS, Fall 2026
Homework 5

Name:
Student ID:

Due on Oct. 25, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to submit your homework in PDF format generated by **LaTeX**, otherwise you will get no points of this homework.
- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1 (20 points)

A/B testing is a randomized experiment used to compare two treatments, Treatment A and Treatment B. Users arrive one by one and are assigned to one of the treatments. Each trial is classified as either a success or a failure.
Consider the following "stay-with-a-winner" procedure:

(i) Assign the first user to Treatment A or Treatment B with equal probabilities.

(ii) If the trial for the nth user is a success, assign the $(n+1)$st user to the same treatment. Otherwise, switch to the other treatment.

The success probabilities for Treatments A and B are a and b, respectively, where $a \neq b$. Given the assigned treatment, these success probabilities do not depend on the previous outcomes.
Let p_n be the probability of success on the nth trial, and let a_n be the probability that Treatment A is assigned on the nth trial.

(a) Show that
$$p_n = (a-b)a_n + b, \qquad a_{n+1} = (a+b-1)a_n + 1 - b.$$

(b) Use part (a) to show that
$$p_{n+1} = (a+b-1)p_n + a + b - 2ab.$$

(c) Assuming that the limit exists, find
$$\lim_{n\to\infty} p_n.$$

Solution.

(a) Let A_n be the event that Treatment A is assigned on trial n, and let S_n be the event that trial n is a success.
$$\Pr(A_n) = a_n, \qquad \Pr(A_n^c) = 1 - a_n.$$
By the law of total probability,
$$\begin{aligned} p_n &= \Pr(S_n \mid A_n)\Pr(A_n) + \Pr(S_n \mid A_n^c)\Pr(A_n^c) \\ &= aa_n + b(1-a_n). \\ \rightarrow \quad p_n &= (a-b)a_n + b. \end{aligned}$$
Treatment A is assigned on trial $n+1$ exactly when trial n uses A and succeeds, or uses B and fails:
$$A_{n+1} = (A_n \cap S_n) \cup (A_n^c \cap S_n^c).$$
The two events on the right are disjoint.
$$\begin{aligned} a_{n+1} &= \Pr(A_n \cap S_n) + \Pr(A_n^c \cap S_n^c) \\ &= aa_n + (1-b)(1-a_n). \\ \rightarrow \quad a_{n+1} &= (a+b-1)a_n + 1 - b. \end{aligned}$$

(b)
$$\begin{aligned} &\because \quad (a-b)a_n = p_n - b, \\ \therefore \quad p_{n+1} &= (a-b)a_{n+1} + b \\ &= (a-b)\big[(a+b-1)a_n + 1 - b\big] + b \\ &= (a+b-1)(p_n - b) + (a-b)(1-b) + b \\ &= (a+b-1)p_n - b(a+b-1) + (a-b)(1-b) + b. \\ -b(a+b-1) &+ (a-b)(1-b) + b = a + b - 2ab. \\ \rightarrow \quad p_{n+1} &= (a+b-1)p_n + a + b - 2ab. \end{aligned}$$

(c) Let

$$L = \lim_{n\to\infty} p_n.$$

$$\because \quad \lim_{n\to\infty} p_{n+1} = L, \qquad \therefore \quad L = (a+b-1)L + a + b - 2ab.$$

$$(2-a-b)L = a + b - 2ab.$$

$$\because \quad 0 \le a, b \le 1, \quad a \ne b, \qquad \therefore \quad 2 - a - b > 0.$$

$$\rightarrow \quad \lim_{n\to\infty} p_n = \frac{a+b-2ab}{2-a-b}.$$

Problem 2 (20 points)

A fair coin is tossed repeatedly and independently, producing a sequence of heads and tails. Let N be the number of tosses required to observe the first occurrence of the pattern HH.
Find the PMF of N.

Solution.
Let

$$p_n = \Pr(N = n).$$

At least two tosses are required, and

$$p_0 = p_1 = 0, \qquad p_2 = \Pr(HH) = \frac{1}{4}.$$

For $n \geq 3$, partition according to the initial pattern:

- T, with probability $1/2$: the process restarts after one toss, and $n-1$ more tosses are required.
- HT, with probability $1/4$: the process restarts after two tosses, and $n-2$ more tosses are required.
- HH, with probability $1/4$: $N = 2$, contributing zero to p_n for $n \geq 3$.

The remaining tosses are independent of the initial pattern.

$$\therefore \quad p_n = \frac{1}{2}p_{n-1} + \frac{1}{4}p_{n-2}, \qquad n \geq 3.$$

Define

$$a_n = 2^n p_n, \qquad n \geq 1.$$

Multiplying the recurrence by 2^n gives

$$a_n = 2^{n-1}p_{n-1} + 2^{n-2}p_{n-2} = a_{n-1} + a_{n-2}.$$

The sequence is specified by

$$a_1 = 0, \qquad a_2 = 1, \qquad a_n = a_{n-1} + a_{n-2} \quad (n \geq 3),$$

with first terms

$$a_1, a_2, a_3, a_4, a_5, a_6, \ldots = 0, 1, 1, 2, 3, 5, \ldots.$$

$$\longrightarrow \quad \Pr(N = n) = \begin{cases} \dfrac{a_n}{2^n}, & n = 2, 3, 4, \ldots, \\ 0, & \text{otherwise.} \end{cases}$$

Problem 3 (15 points)

A system contains seven devices: four in Group A and three in Group B. Each device fails independently with probability p, where $0 < p < 1$.
Let X be the number of failed devices in Group A, and let T be the total number of failed devices in the system.

(a) Find the PMF of X and the PMF of T.

(b) Given that exactly three devices have failed, find the conditional PMF of X. Identify the distribution.

(c) Explain why the conditional distribution in part (b) does not depend on p.

Solution.

(a) The device failures are independent, each with probability p.

$$\therefore \quad X \sim \text{Binomial}(4, p), \qquad T \sim \text{Binomial}(7, p).$$

Their PMFs are

$$P(X = x) = \begin{cases} \binom{4}{x} p^x (1-p)^{4-x}, & x = 0, 1, 2, 3, 4, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$P(T = t) = \begin{cases} \binom{7}{t} p^t (1-p)^{7-t}, & t = 0, 1, \ldots, 7, \\ 0, & \text{otherwise.} \end{cases}$$

(b) For $x = 0, 1, 2, 3$, the event $\{X = x, T = 3\}$ requires x failures in Group A and $3 - x$ failures in Group B. By independence,

$$\begin{aligned} P(X = x, T = 3) &= \binom{4}{x} p^x (1-p)^{4-x} \binom{3}{3-x} p^{3-x} (1-p)^x \\ &= \binom{4}{x} \binom{3}{3-x} p^3 (1-p)^4. \end{aligned}$$

Also,

$$P(T = 3) = \binom{7}{3} p^3 (1-p)^4.$$

$$\begin{aligned} P(X = x \mid T = 3) &= \frac{P(X = x, T = 3)}{P(T = 3)} \\ &= \frac{\binom{4}{x}\binom{3}{3-x} p^3 (1-p)^4}{\binom{7}{3} p^3 (1-p)^4} \\ &= \frac{\binom{4}{x}\binom{3}{3-x}}{\binom{7}{3}}. \end{aligned}$$

$$\longrightarrow \quad P(X = x \mid T = 3) = \begin{cases} \dfrac{\binom{4}{x}\binom{3}{3-x}}{35}, & x = 0, 1, 2, 3, \\ 0, & \text{otherwise.} \end{cases}$$

Explicitly,

x	0	1	2	3
$P(X = x \mid T = 3)$	$\frac{1}{35}$	$\frac{12}{35}$	$\frac{18}{35}$	$\frac{4}{35}$

Problem 3 continued on next page...

This is a hypergeometric distribution:

$$X \mid (T = 3) \sim \text{Hypergeometric}(N = 7, K = 4, n = 3),$$

corresponding to selecting 3 devices without replacement from 7 devices, of which 4 belong to Group A.

(c) Every specified configuration with exactly three failed devices and four working devices has probability

$$p^3(1-p)^4.$$

Its conditional probability given $T = 3$ is

$$\frac{p^3(1-p)^4}{\binom{7}{3}p^3(1-p)^4} = \frac{1}{35}.$$

All $\binom{7}{3} = 35$ such configurations are equally likely after conditioning on $T = 3$. The common factor $p^3(1-p)^4$ cancels:

$$\longrightarrow \quad P(X = x \mid T = 3) = \frac{\binom{4}{x}\binom{3}{3-x}}{\binom{7}{3}}, \qquad x = 0, 1, 2, 3,$$

which does not depend on p.

Problem 4 (15 points)

Two fair six-sided dice are rolled independently. Let U and V be their respective outcomes, and define

$$D = |U - V|.$$

(a) Find the PMF of D.

(b) Find the CDF $F_D(t)$ for every real number t.

(c) Define $W = \min(D, 2)$. Find the PMF of W.

Solution.
The 36 ordered outcomes are equally likely:

$$P(U = u, V = v) = P(U = u)P(V = v) = \frac{1}{36}, \qquad u, v \in \{1, \ldots, 6\}.$$

(a) The possible values of D are $0, 1, \ldots, 5$.

For $d = 0$,

$$P(D = 0) = \sum_{k=1}^{6} P(U = k, V = k) = \frac{6}{36} = \frac{1}{6}.$$

For $d \in \{1, \ldots, 5\}$,

$$\begin{aligned} P(D = d) &= P(U - V = d) + P(V - U = d) \\ &= \sum_{k=1}^{6-d} [P(U = k + d, V = k) + P(U = k, V = k + d)] \\ &= \frac{2(6 - d)}{36} = \frac{6 - d}{18}. \end{aligned}$$

$$\rightarrow \quad p_D(d) = \begin{cases} \frac{1}{6}, & d = 0, \\ \frac{6-d}{18}, & d \in \{1, 2, 3, 4, 5\}, \\ 0, & \text{otherwise.} \end{cases}$$

(b) By definition,

$$F_D(t) = P(D \le t) = \sum_{\substack{d \in \{0,1,\ldots,5\} \\ d \le t}} p_D(d).$$

The cumulative probabilities are

$$\begin{aligned} P(D \le 0) &= \frac{6}{36} = \frac{1}{6}, \\ P(D \le 1) &= \frac{6 + 10}{36} = \frac{4}{9}, \\ P(D \le 2) &= \frac{6 + 10 + 8}{36} = \frac{2}{3}, \\ P(D \le 3) &= \frac{6 + 10 + 8 + 6}{36} = \frac{5}{6}, \\ P(D \le 4) &= \frac{6 + 10 + 8 + 6 + 4}{36} = \frac{17}{18}, \\ P(D \le 5) &= 1. \end{aligned}$$

$$\rightarrow \quad F_D(t) = \begin{cases} 0, & t < 0, \\ \frac{1}{6}, & 0 \le t < 1, \\ \frac{4}{9}, & 1 \le t < 2, \\ \frac{2}{3}, & 2 \le t < 3, \\ \frac{5}{6}, & 3 \le t < 4, \\ \frac{17}{18}, & 4 \le t < 5, \\ 1, & t \ge 5. \end{cases}$$

(c) The definition of W gives

$$W = \begin{cases} 0, & D = 0, \\ 1, & D = 1, \\ 2, & D \ge 2. \end{cases}$$

$$\begin{aligned} P(W = 0) &= P(D = 0) = \frac{1}{6}, \\ P(W = 1) &= P(D = 1) = \frac{5}{18}, \\ P(W = 2) &= P(D \ge 2) \\ &= 1 - P(D = 0) - P(D = 1) \\ &= 1 - \frac{1}{6} - \frac{5}{18} = \frac{5}{9}. \end{aligned}$$

$$\rightarrow \quad p_W(w) = \begin{cases} \frac{1}{6}, & w = 0, \\ \frac{5}{18}, & w = 1, \\ \frac{5}{9}, & w = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Problem 5 (15 points)

There are two coins, A and B. Coin A lands heads with probability 1/4, while coin B lands heads with probability 3/4.
One coin is selected with equal probabilities and then tossed twice. The same selected coin is used for both tosses. Given the selected coin, the two tosses are independent.
Let $C \in \{A, B\}$ denote the selected coin. Let X and Y be the indicator random variables for heads on the first and second tosses, respectively.

(a) Find the PMF of X and the PMF of Y.

(b) Are X and Y independent? Are they conditionally independent given C? Justify your answers.

(c) Given that both tosses are heads, find the conditional probability that coin B was selected.

Solution.

(a) By the law of total probability,

$$\begin{aligned} P(X=1) &= P(C=A)P(X=1 \mid C=A) \\ &\quad + P(C=B)P(X=1 \mid C=B) \\ &= \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{3}{4} = \frac{1}{2}. \end{aligned}$$

Similarly,

$$P(Y=1) = \frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{3}{4} = \frac{1}{2}.$$

Also,

$$P(X=0) = 1 - P(X=1) = \frac{1}{2}, \qquad P(Y=0) = 1 - P(Y=1) = \frac{1}{2}.$$

$$\implies \quad p_X(k) = p_Y(k) = \begin{cases} \frac{1}{2}, & k \in \{0,1\}, \\ 0, & \text{otherwise.} \end{cases}$$

(b) Using conditional independence given the selected coin,

$$\begin{aligned} P(X=1,Y=1) &= P(C=A)P(X=1,Y=1 \mid C=A) \\ &\quad + P(C=B)P(X=1,Y=1 \mid C=B) \\ &= \frac{1}{2}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2 \\ &= \frac{1}{32} + \frac{9}{32} = \frac{5}{16}. \end{aligned}$$

However,

$$P(X=1)P(Y=1) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}.$$

$$P(X=1,Y=1) \neq P(X=1)P(Y=1) \quad \implies \quad X \text{ and } Y \text{ are not independent.}$$

For conditional independence, write

$$p_A = \frac{1}{4}, \qquad p_B = \frac{3}{4}.$$

The problem states that the two tosses are independent for each fixed selected coin. For every $c \in \{A, B\}$ and $x, y \in \{0, 1\}$,

$$\begin{aligned} P(X=x, Y=y \mid C=c) &= p_c^{x+y}(1-p_c)^{2-x-y} \\ &= \left[p_c^x(1-p_c)^{1-x}\right]\left[p_c^y(1-p_c)^{1-y}\right] \\ &= P(X=x \mid C=c)\, P(Y=y \mid C=c). \end{aligned}$$

$$\implies \quad X \text{ and } Y \text{ are conditionally independent given } C.$$

Problem 5 continued on next page...

(c) By Bayes' formula,

$$\begin{aligned} P(C = B \mid X = 1, Y = 1) &= \frac{P(C = B)\, P(X = 1, Y = 1 \mid C = B)}{P(X = 1, Y = 1)} \\ &= \frac{\frac{1}{2}\left(\frac{3}{4}\right)^2}{\frac{1}{2}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2} \\ &= \frac{9/32}{10/32}. \end{aligned}$$

$$\implies \quad P(C = B \mid X = 1, Y = 1) = \frac{9}{10}.$$

Problem 6 (15 points)

A random variable X has PMF

$$\Pr(X=1)=\frac{1}{2},\qquad \Pr(X=2)=\frac{1}{4},\qquad \Pr(X=3)=\Pr(X=4)=\frac{1}{8}.$$

Let U have the discrete uniform distribution on $\{1,2,3,4\}$.
Throughout this problem, logarithms have base 2, and entropy is measured in bits.

(a) Find $H(X)$ and $H(U)$.

(b) Define

$$Y=\begin{cases}1, & X\le 2,\\ 0, & X>2,\end{cases}\qquad Z=\begin{cases}1, & X=1,\\ 0, & X\neq 1.\end{cases}$$

Find the PMF of each variable and calculate $H(Y)$ and $H(Z)$.

(c) Is the size of the support sufficient to determine the entropy of a discrete random variable? Explain using distributions from this problem.

Solution.

(a) By the definition of entropy,

$$H(X)=-\sum_x \Pr(X=x)\log_2\Pr(X=x).$$

$$\begin{aligned}H(X)&=-\left(\frac{1}{2}\log_2\frac{1}{2}+\frac{1}{4}\log_2\frac{1}{4}+\frac{1}{8}\log_2\frac{1}{8}+\frac{1}{8}\log_2\frac{1}{8}\right)\\&=\frac{1}{2}+\frac{1}{2}+\frac{3}{8}+\frac{3}{8}=\frac{7}{4}.\end{aligned}$$

For U,

$$\Pr(U=u)=\frac{1}{4},\qquad u\in\{1,2,3,4\}.$$

$$\rightarrow\quad H(U)=-\sum_{u=1}^{4}\frac{1}{4}\log_2\frac{1}{4}=2.$$

(b) For Y,

$$\begin{aligned}\Pr(Y=1)&=\Pr(X\le 2)=\frac{1}{2}+\frac{1}{4}=\frac{3}{4},\\ \Pr(Y=0)&=\Pr(X>2)=\frac{1}{8}+\frac{1}{8}=\frac{1}{4}.\end{aligned}$$

$$\rightarrow\quad p_Y(y)=\begin{cases}\frac{3}{4}, & y=1,\\ \frac{1}{4}, & y=0,\\ 0, & \text{otherwise.}\end{cases}$$

$$\begin{aligned}H(Y)&=-\frac{3}{4}\log_2\frac{3}{4}-\frac{1}{4}\log_2\frac{1}{4}\\&=-\frac{3}{4}(\log_2 3-2)+\frac{1}{2}\\&=2-\frac{3}{4}\log_2 3\approx 0.8113.\end{aligned}$$

For Z,

$$\begin{aligned}\Pr(Z=1)&=\Pr(X=1)=\frac{1}{2},\\ \Pr(Z=0)&=\Pr(X\neq 1)=1-\frac{1}{2}=\frac{1}{2}.\end{aligned}$$

Problem 6 continued on next page...

$$\rightarrow \quad p_Z(z) = \begin{cases} \frac{1}{2}, & z \in \{0,1\}, \\ 0, & \text{otherwise.} \end{cases}$$

$$H(Z) = -\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{2}\log_2\frac{1}{2} = 1.$$

(c) No. The support size alone does not determine entropy.

$$\text{supp}(X) = \text{supp}(U) = \{1,2,3,4\}, \qquad H(X) = \frac{7}{4} \neq 2 = H(U).$$

Another example is

$$\text{supp}(Y) = \text{supp}(Z) = \{0,1\}, \qquad H(Y) = 2 - \frac{3}{4}\log_2 3 \neq 1 = H(Z).$$

$\rightarrow$ Entropy also depends on the probabilities assigned to the support values.